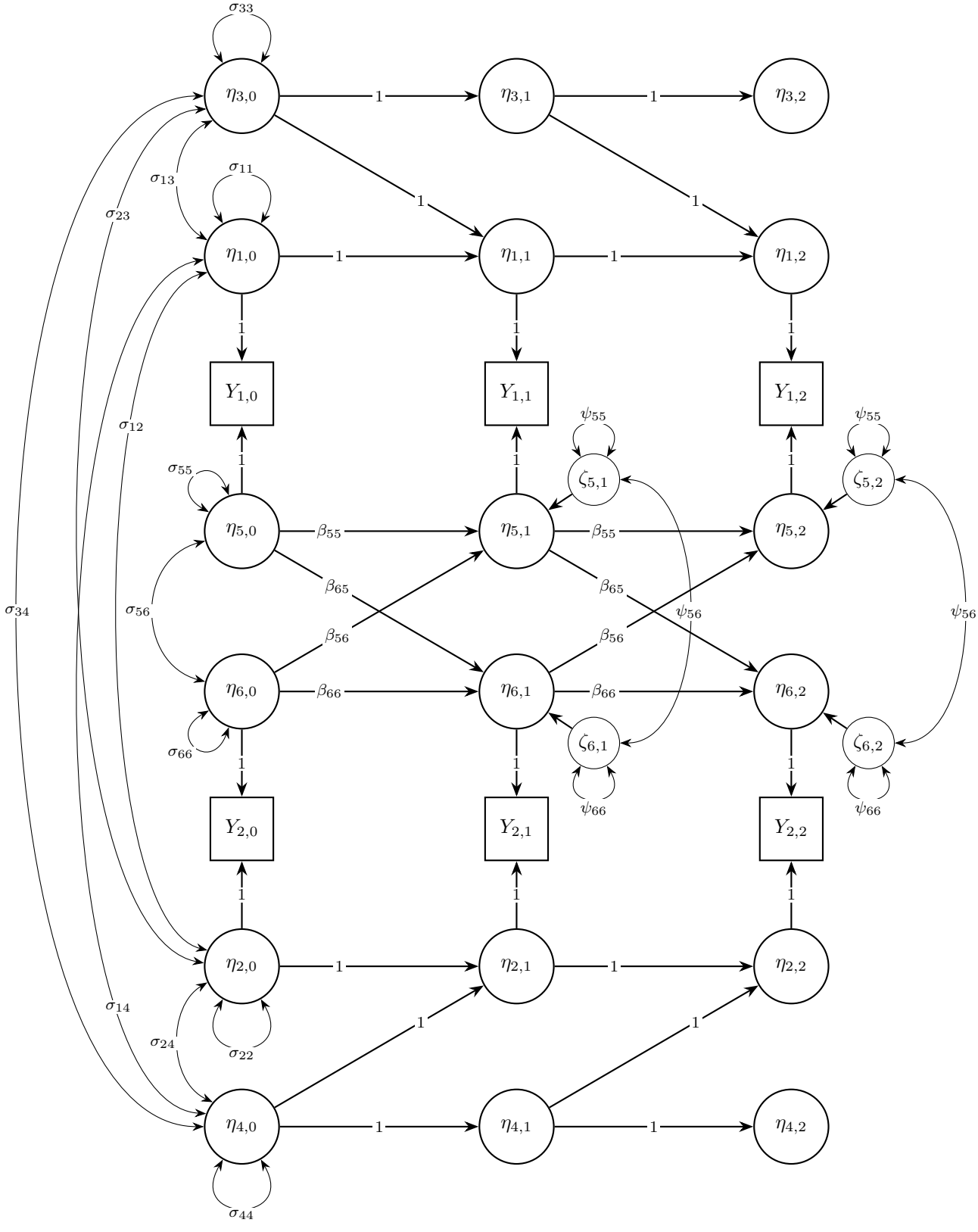


$$\boldsymbol{\eta}_t = (\eta_{1,t}, \ldots, \eta_{6,t})', \quad \mathbf{y}_t = (Y_{1,t}, Y_{2,t})'; \text{ no measurement error}$$

$$\text{levels } \eta_{1,t}, \eta_{2,t}; \text{ slopes } \eta_{3,t} = \eta_3,\ \eta_{4,t} = \eta_4 \text{ for all } t; \text{ structured residuals } \eta_{5,t}, \eta_{6,t}$$

$$\eta_{1,t} = \eta_{1,0} + t\eta_3, \quad \eta_{2,t} = \eta_{2,0} + t\eta_4, \quad Y_{1,t} = \eta_{1,t} + \eta_{5,t}, \quad Y_{2,t} = \eta_{2,t} + \eta_{6,t}$$

$$\boldsymbol{\beta} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \beta_{55} & \beta_{56} \\ 0 & 0 & 0 & 0 & \beta_{65} & \beta_{66} \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \qquad \boldsymbol{\Psi} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Psi}_{\text{SR}} \end{pmatrix},\ \boldsymbol{\Psi}_{\text{SR}} = \begin{pmatrix} \psi_{55} & \psi_{56} \\ \psi_{65} & \psi_{66} \end{pmatrix}$$

$$\boldsymbol{\eta}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad \boldsymbol{\mu}_0 = (\mu_1, \mu_2, \mu_3, \mu_4, 0, 0)', \quad \boldsymbol{\Sigma}_0 = \begin{pmatrix} \boldsymbol{\Sigma}_{\text{G}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{\text{SR}} \end{pmatrix}$$

$$\boldsymbol{\Sigma}_{\text{G}} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \sigma_{34} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_{44} \end{pmatrix}, \quad \boldsymbol{\Sigma}_{\text{SR}} = \begin{pmatrix} \sigma_{55} & \sigma_{56} \\ \sigma_{65} & \sigma_{66} \end{pmatrix}$$